

ESA KHAN

Machine Learning Engineer — LLM Systems in Production · RAG · Agentic AI · Generative AI

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AI/ML Engineer with hands-on, production experience across Machine Learning, Deep Learning, and NLP — building and integrating LLM-based solutions (RAG, agentic pipelines, fine-tuning) and AI-driven automation into real applications.

EDUCATION

Bachelor of Computer Science — COMSATS University Islamabad, Abbottabad Campus

2026

FSc Pre-Engineering / Computer Science — Govt Postgraduate College, Lakki Marwat

2022

EXPERIENCE

AI/ML Engineer — Tekjee (US-based company) — Part-time, Onsite

Present

- Work hands-on across every level of Retrieval-Augmented Generation — from naive/basic RAG to advanced and agentic RAG pipelines — covering chunking, embeddings, retrieval, and re-ranking.
- Build agentic AI workflows and multi-step tool-using agents with LangChain and LangGraph.
- Fine-tune and adapt LLMs, and integrate LLM-based solutions (multiple provider APIs — API key setup, auth, and usage management) into production services.
- Develop and expose AI services through FastAPI, and deploy trained/served models on AWS.

SELECTED PROJECTS

AI Gig Generator & Resume-Based LinkedIn Job Matcher *RAG · Python/FastAPI · KNN · Cosine Similarity · LangChain/LangGraph · FAISS*

- Own the AI/ML module as a standalone Python/FastAPI microservice that plugs into a Node.js/React application through a defined JSON handoff contract — mirrors a real service-integration pattern for shipping AI features into an existing product.
- Built the resume-upload-to-job-matching pipeline: embeddings, KNN retrieval, and cosine-similarity re-ranking against live LinkedIn job postings.
- Added LangGraph agentic loops that iteratively refine generated gig content based on skill level, niche, and user feedback.

AI-Powered Self-Correcting Code Assistant *Python · LangChain · Groq API · Streamlit*

- Built a coding assistant that detects and fixes errors through iterative LLM reasoning loops, using a chain-of-thought self-correction pipeline with feedback, cutting debugging time significantly.
- Exposed the assistant through a Streamlit interface backed by prompt-engineered, structured LLM calls.

AI Video Agent — Audio Understanding & RAG-Based Q&A *LangChain · Whisper · Groq API · FAISS*

- Built a multimodal RAG agent: Whisper transcribes video audio, FAISS-indexed embeddings power low-latency retrieval, and an LLM answers natural-language questions over long transcripts.
- Delivered through a real-time Streamlit interface for interactive Q&A.

AI Attendance System — Voice & Face Recognition *Deep Learning · OpenCV · FaceNet*

- Built a dual-modal, contactless attendance system combining FaceNet/DeepFace face recognition with speech-based voice verification, logging attendance on a successful match.

SKILLS

ML / DL Libraries: TensorFlow, PyTorch, Scikit-learn, Keras, NumPy, Pandas, OpenCV

NLP & Transformers: Hugging Face Transformers, BERT/GPT-style architectures, tokenization, sentence-transformers

LLM & Generative AI: OpenAI API, Groq API, prompt engineering, structured output generation, function/tool calling, chain-of-thought reasoning, API-key-based provider integration

RAG & Retrieval: Hands-on across every level — naive, advanced, and agentic RAG; chunking strategies, embeddings, re-ranking (KNN, cosine similarity), vector search

Agentic AI: LangChain, LangGraph, multi-step agent design, tool-using agents, self-correcting/reflection loops, memory & state management

AI-Driven Automation: n8n workflow automation, agent orchestration, API integrations, webhook-driven pipelines

Model Fine-Tuning & Serving: Fine-tuning open-source LLMs, LoRA/PEFT, dataset preparation, local model serving with Ollama

Vector Databases: FAISS, Chroma

Programming: Python (Advanced — OOP, scripting, automation), SQL

Backend & APIs: FastAPI, Flask, Streamlit — building and integrating AI services with APIs and databases

MLOps & Cloud: Model deployment on AWS, versioning, monitoring and experiment tracking for ML/LLM pipelines; Docker (basic)

Tools & Platforms: Git, GitHub, VS Code, Jupyter Notebook, Linux, MCP

CERTIFICATIONS & ACHIEVEMENTS

- Huawei HCIA-AI V3.5 (x2) — 2024
- Generative AI with AWS
- Generative AI Specialization — DeepLearning.AI (Coursera)
- Deep Learning Unlocked
- Deep Learning Workshop Certificate — COMSATS University
- Campus Honor Roll — Academic Excellence
- Open Source Contributor — github.com/esakhan786